CashJournal — Complete Application Audit & Feature Reference

Comprehensive architectural, mathematical, UI/UX, localization, and security audit of the native Android codebase (v1.0.14-dev).

Audit Scope & Verification: Covers 100% of Kotlin source files, Gradle build scripts, Room SQLite schemas (v2), DataStore preferences, domain algorithms, 7 primary Compose screens, 8 modal bottom sheets, sensory engines, and network/backup security configs. All 9 JUnit4 test suites (testDebugUnitTest) executed and passed cleanly.

1. Executive Summary & Architectural Overview

CashJournal (app.cashjournal) is a native, offline-first personal finance and cash-flow telemetry application for Android (minSdk 26 / Android 8.0 through targetSdk 34 / Android 14). Built with Kotlin 2.0, Jetpack Compose, Room SQLite v2, Preferences DataStore, Kotlinx Serialization, OkHttp 4.12, and Phosphor Icons, the app is engineered around five non-negotiable pillars:

- 100% Free & Keyless: Zero paid libraries, zero user accounts, and zero API keys. Weather and geocoding use Open-Meteo; daily exchange rates use ExchangeRate-API Open Access.
- 100% Local & Private: Financial data never leaves the device sandbox. Android Auto Backup and cloud/device-transfer extraction are disabled via allowBackup="false" and explicit XML exclusion rules.
- Zero-Drift Mathematical Integrity: All currency values are stored as 64-bit integer minor units (Long). Account balances are dynamically derived in SQLite from initialBalance + SUM(inflows) - SUM(outflows).
- Dual-Layer Network Lockdown: Both Android's network_security_config.xml and an OkHttp HostAllowlistInterceptor restrict outbound HTTPS requests to exactly three domains: api.open-meteo.com, geocoding-api.open-meteo.com, and open.er-api.com.
- Ergonomic One-Handed UI & Sensory Feedback: Spring-animated floating SideRail that swaps between left and right edges via UI toggle or physical device knock/shake (ShakeDetector), paired with procedural 44.1kHz PCM audio chimes (ChimeGenerator) and GPU-accelerated background blobs (AmbientField).

2. Complete Feature & Functional Reference

2.1 Entry Point, App Lock & Lifecycle (MainActivity.kt)

- Edge-to-Edge Portrait Shell: Extends FragmentActivity (for BiometricPrompt compatibility) and invokes enableEdgeToEdge().
- ADB Intent Seeding: Passing --ez seed true via launch intent triggers viewModel.seedExampleData().
- Screen Privacy (FLAG_SECURE): Reactively sets or clears WindowManager.LayoutParams.FLAG_SECURE when Screen Privacy is toggled, blocking screenshots and screen recording.
- 15-Second Background Auto-Lock: Records lastBackgroundTimestamp in onStop(); if backgrounded for >15,000 ms, onStart() re-engages AppLockScreen when PIN protection is enabled.

2.2 First-Run Onboarding Flow (OnboardingScreen.kt — 8 Steps)

Step	Title	Functional Behavior & Controls
Step 1	Welcome	Displays official squircle emblem and 3 privacy guarantee tiles (ShieldCheck: No account, DeviceMobile: Stays on phone, CloudSlash: Nothing uploaded).
Step 2	Language	Instant switch between English and Bahasa Indonesia with 250ms crossfade.

Step	Title	Functional Behavior & Controls
Step 3	Region & Currency	Searchable Country/Currency list with vector CountryFlag icons and currency code badges (IDR, USD, EUR, GBP, SGD, JPY, AUD, MYR).
Step 4	Name	Optional 40-character name input with live SpaceGrotesk counter (N/40) and Skip action.
Step 5	Weather Privacy	Choose Approximate Location (~1 km rounded), Search City (8 popular city chips + custom query), or Not Now.
Step 6	Permissions	Interactive runtime permission cards for Coarse Location and Notifications (API 33+), plus Device Sandbox storage transparency card.
Step 7	Menu Layout	Left/Right SegmentedControl with live-animating mini phone wireframe preview and Shake to Swap toggle.
Step 8	Start Mode	Select Load Demo Mode (4 accounts, 10 transactions, 7.5M IDR budget) or Start from scratch, followed by CheckCircle completion screen.

2.3 Floating SideRail, Status Cluster & Shared Components

- Spring Physics Side Swap: Content margins mirror between 20.dp and 78.dp while the floating rail leaps across the screen with a ±18° tilt angle and haptic tick.
- Unified Status Ring (RailStatusCluster.kt): 42.dp Canvas gauge rendering a 270° live battery arc (Emerald >30%, Saffron 11–30%, Berry <=10%), 4 cellular signal dots (normalized from 4-bar and 5-bar OEM scales at 60°, 80°, 100°, 120° via SignalStrengthHelper), a center Eye/EyeSlash privacy toggle, and a live HH:mm clock. Morphs into a CaretLeft back button on AccountDetailScreen.
- Liquid Tab Drip & Tooltips: Switching tabs animates a stretching Cobalt blob over 420ms. Long-pressing any rail icon displays an inward frosted tooltip pill with shockwave animation on side swap.
- Per-Digit Wave Privacy Masking (MaskedAmount.kt): Hiding/revealing balances triggers a 32ms/digit wave animation converting numbers into 7 dots (•••••••) while keeping hero currency symbols visible.
- Ambient Field (AmbientField.kt): Up to 6 account-colored radial blobs scaled by (balance / maxBalance)^0.8 drifting on a 30-second sinusoidal loop in isolated GPU layers.

2.4 Primary Screens (7 Screens)

Screen	Key Features, Telemetry & Interactive Flows
DashboardScreen	Total Net Worth hero header; 28.dp Safe-to-Spend Today glass card with 64.dp circular % spent gauge and status badge (Safe, Near Limit, Recalibrate, Set Budget, No Funds); Active Accounts horizontal carousel (130x96.dp cards with radial accent haze); Recent Activity (top 5 transactions); Monthly Budget Runway card; bottom mist gradient and ScrollForMorePill.
LedgerScreen	Activity Ledger with 4 live-count filter pills (All, Expense, Income, Transfer); instant search bar matching notes, EN/ID category names, and raw amount digits; chronological day grouping (TODAY, YESTERDAY, weekday/date headers); tap any row to open TransactionEditSheet.
AccountsScreen	World Currencies card with converted Net Worth, 1-unit cross-rate, online/offline status dot, manual refresh, 8 flag chips (USD, EUR, SGD, JPY, GBP, MYR, SAR, THB), and '+ World' button; 24.dp account glass cards with 4.dp left accent bar, % of total pill, and relative balance bar; interactive Reorder Mode (ArrowUp/ArrowDown).
AccountDetailScreen	Single-account detail view with % of total badge, 24.dp balance card, Reconcile Balance button, side-by-side Inflow (Emerald: income + transfers in) and Outflow (Charcoal: expense + transfers out) tiles, and filtered account transaction history.
AnalyticsScreen	Last 6 Months Inflow/Outflow paired bar chart (transfers excluded); interactive Monthly Spending Runway calibrator with - / Slider / + step buttons (500k IDR steps); cycle-filtered Top Spending Categories Donut Chart (top 4 categories + Other).

Screen	Key Features, Telemetry & Interactive Flows
SettingsScreen	8 configuration cards: Language (EN/ID), Main Currency, Payday Cycle Start Day (1..28 slider dialog with live cycle preview), Security & Access (PIN Lock, Biometric Unlock, FLAG_SECURE Screen Privacy, Lock App Now), Sensory & Feedback (Sound, Haptics, Daily Reminder), Sidebar Placement & Knock Sensitivity (Gentle/Balanced/Firm), Weather & °C/°F, and Data & Portability (SAF JSON Export/Restore, Replay Onboarding, Clear All Data, About).
AppLockScreen	Full-screen security gate with 4 spring-animated PIN dots, horizontal error shake + reject haptics, 3x4 numeric keypad, and automatic BiometricPrompt trigger (300ms after launch).

2.5 Modal Bottom Sheets (8 Sheets)

Sheet	Capabilities & Validation Rules
LogTransactionSheet	Segmented type selector (Expense/Income/Transfer); backdating chips (Today, Yesterday, Custom DatePickerDialog); source/destination account chips; note input with real-time keyword category detection + manual category picker dialog; 36.sp amount display; CustomNumericKeypad (000 for zero-decimal, 00 for 2-decimal); 14-droplet SaveSplash burst.
TransactionEditSheet	Retroactive editing of all transaction fields (type, date, source/destination accounts, note, category, amount) and safe deletion with confirmation dialog; triggers atomic balance recalculation across all old and new affected accounts.
ReconcileBalanceSheet	Displays current app balance and live adjustment delta (+ Emerald / - Persimmon) as the user enters their actual real-world balance; creates an audited INCOME or EXPENSE adjustment entry.
NewAccountSheet	Creates a new account with name (max 28 chars), starting initialBalance, and 6 AccentKey color swatches.
MainCurrencySheet	Searchable country/currency list with confirmation dialog that converts all accounts, transactions, and budgets at the cached exchange rate.
WorldCurrenciesSheet	Searchable list of 16 world currencies with vector flags, sync status badge, 1-unit exchange rate, and converted Net Worth preview.
WeatherDetailSheet	Switches between Approximate Location (~1 km rounded), City Search (live Open-Meteo geocoding), and Off.
PinSetupSheet	Contains both PinSetupSheet (2-step create & confirm 4-digit PIN) and PinVerifyOrRemoveSheet (verifies current PIN before removal).

3. Data Architecture, Schemas & Mathematical Engine

3.1 Room Database (cashjournal_db v2) & Zero-Drift SQL Formula

The Room database defines three tables: accounts (id, name, accentKey, balance, initialBalance, sortIndex, updatedAt), transactions (id, title, category, type, amount, date, iconKey, accentKey, sourceAccountId [indexed], destinationAccountId [indexed], updatedAt), and monthly_plans (id='primary_plan', budget, startingSpent, monthKey). Every account's balance is derived via SQL:

```
balance = initialBalance + SUM(INCOME where src=id) + SUM(TRANSFER where dst=id) - SUM(EXPENSE where src=id) - SUM(TRANSFER where src=id)
```

3.2 Safe-to-Spend & Custom Cycle Calculator (SafeToSpendCalculator.kt)

- Custom Billing Cycle (cycleStartDay in 1..28): Computes exact cycle start (00:00:00.000) and end (23:59:59.999) timestamps, date range labels (e.g. '25 Sep – 24 Oct 2026'), and remaining days clamped to >= 1.
- Daily Safe-to-Spend Formula: perDay = floor(max(0, budget - totalSpent) / max(1, daysLeft)), where totalSpent = startingSpent + SUM(cycleExpenses). Internal transfers are strictly excluded.
- Status Priority: NO_FUNDS (no accounts or netWorth <= 0) -> SET_BUDGET (budget <= 0) -> RECALIBRATE (remaining <= 0) -> NEAR_LIMIT (spent/budget >= 70%) -> SAFE.

3.3 Exchange Rate & Minor-Unit Conversion Math (CurrencyConverter.kt)

Supports 16 currencies (zero-decimal: IDR, JPY, KRW; two-decimal: USD, EUR, GBP, SGD, AUD, MYR, SAR, CNY, CAD, CHF, HKD, NZD, THB). Auto-refreshes at most every 12 hours; manual refresh rate-limited to once per hour:

```
targetMinor = round( (amountMinor / 10^sourceDecimals) * (rateTarget / rateSource) * 10^targetDecimals )
```

4. Automated Unit Test Suite Status

All 9 JVM JUnit4 test classes in app/src/test/java/app/cashjournal/ passed cleanly during audit verification:

Test Suite	Verified Assertions & Coverage	Result
AcceptanceCheckTest	Build Brief Section 16 acceptance numbers (Rp 18.45M net worth, Rp 150.294/day on Oct 14, 66% ring, Rp 1M expense -> Rp 91.470/day & 79% Near Limit, transfer isolation, knock sensitivity, 7-dot mask).	**PASS**
LedgerIntegrityTest	Zero-drift balance derivation on retroactive edits/deletes, atomic transfer net-worth conservation, manual balance reconciliation delta math, and backdating.	**PASS**
SafeToSpendTest	Leap-year February handling, status thresholds, last-day clamping, custom cycleStartDay (1, 25, boundary clamping 0 & 31), and cross-month cycle expense filtering.	**PASS**
BackupRestoreTest	BackupPayload JSON round-trip serialization, corrupted JSON failure handling, version <= 0 rejection, and zero-drift recalculation after restore.	**PASS**
CurrencyConverterTest	Same-currency identity, USD <-> IDR decimal scaling (2 vs 0 decimals), EUR -> JPY cross-rates, and missing-rate fallback.	**PASS**
CategoryDetectorTest	Keyword detection across FOOD, TRANSPORT, SHOPPING, UTILITIES, HOUSING, INVESTMENT, SALARY, and OTHER.	**PASS**
HostAllowlistTest	Verifies HostAllowlistInterceptor permits the 3 allowed hosts and throws SecurityException on unauthorized domains.	**PASS**
SignalStrengthHelperTest	Monotonic translation of 4-bar and 5-bar cellular signal scales into 4 UI dots.	**PASS**
WeatherIconMapperTest	WMO weather code mapping across day/night variants to Phosphor icons.	**PASS**

5. Comprehensive Audit Findings: Bugs, Edge Cases & Risks

CRITICAL FINDING (CRITICAL-1 — MainViewModel.kt:708-719): Changing the main currency converts AccountEntity.balance but omits AccountEntity.initialBalance. Because AccountDao.recalculateAccountBalance() recomputes balance = initialBalance + SUM(net_flows) whenever a transaction is logged, edited, or deleted, the account balance will suddenly revert its base back to the unconverted old currency amount on the very next transaction!

ID	Severity	Location	Audit Finding & Impact
CRITICAL-1	Critical	MainViewModel.kt:708-719	changeMainCurrency() converts acc.balance but forgets acc.initialBalance, corrupting account balances on the next transaction recalculation.
HIGH-1	High	MainViewModel.kt:109,158,250	_selectedAccount stores a static snapshot; AccountDetailScreen balance card and % of total do not update after editing/deleting transactions or reconciling balance.
HIGH-2	High	AccountsScreen.kt:88-119	AccountsScreen uses local remember { mutableStateOf('USD') } instead of state.userPreferences.selectedWorldCurrency, ignoring WorldCurrenciesSheet selections. Also divides by hardcoded IDR rates when cachedRates is null.
HIGH-3	High	MainAppScaffold.kt:197-240	Selecting 'Not now' / 'Off' for weather never hides the weather widget in GreetingBar; it permanently displays a fake 24°C (75°F) Sun icon.
HIGH-4	High	SafeToSpendCalculator.kt:132-137	MonthlyPlan.startingSpent checks plan.monthKey == todayMonthKey (calendar month), causing startingSpent to drop to 0 mid-cycle when cycleStartDay > 1 spans two calendar months.
MED-1	Medium	Category.kt:32-102,137-148	Substring matching (lower.contains(keyword)) causes false positives on short keywords ('theater'/'sweater' match 'eat' -> FOOD; 'repair'/'chair' match 'air' -> UTILITIES).
MED-2	Medium	AnalyticsScreen.kt:443-454	Dragging the budget Slider invokes viewModel.updateBudget() on every frame, spawning dozens of concurrent AudioTrack chimes and Room DB writes.
MED-3	Medium	LogTransactionSheet.kt:204 / TransactionEditSheet.kt:471	(a) SALARY is filtered out of LogTransactionSheet even for INCOME; (b) typing a note in TransactionEditSheet overwrites manual category selection; (c) editTransaction overwrites expense category accent with account accent.
MED-4	Medium	UserPreferencesRepository.kt:303 / build.gradle.kts:55	4-digit PIN is hashed with unsalted SHA-256; release build type uses debug.keystore and isMinifyEnabled = false.
MED-5	Medium	Multiple UI & ViewModel files	Strings in GreetingBar, DashboardScreen, AccountsScreen, AccountDetailScreen, AnalyticsScreen, sheets, and ViewModel toasts remain hardcoded in English when Bahasa Indonesia ('in') is selected.
LOW-1	Low	BackupManager.kt:133-145	PreferencesBackupDto omits screenPrivacyEnabled, weatherOption, weatherCityLabel, weatherCityLat, weatherCityLon, and selectedWorldCurrency.
LOW-2	Low	SettingsScreen.kt:624 / OnboardingScreen.kt:1171	Onboarding requests POST_NOTIFICATIONS and Settings toggles dailyReminderEnabled, but no AlarmManager/WorkManager scheduler is implemented.
LOW-3	Low	Multiple UI Components	tactileHapticsEnabled preference is bypassed in SegmentedControl, SettingToggleRow, CustomNumericKeypad call sites, OnboardingScreen, and PinSetupSheet.

ID	Severity	Location	Audit Finding & Impact
LOW-4	Low	MainViewModel.kt:564 / SettingsScreen.kt:532	Turning Sound Effects off plays a chime before the preference updates; PIN/Biometric actions trigger duplicate toast calls.
LOW-5	Low	LedgerScreen.kt:143 / SafeToSpendCalculator.kt:90	'Yesterday' check uses nowYear == selYear && nowDay - 1 == selDay (fails on Jan 1 for Dec 31); diffMillis / 86_400_000L integer division truncates 1 day across DST spring-forward transitions.
LOW-6	Low	build.gradle.kts:11 / ui/preview/* / AppStrings.kt	Unused gitCommitCount process in Gradle, unreferenced preview screens, unused onPinChange callback, and 9 unused localized string properties.

5.1 Recommended Remediation Plan

1. Fix Main Currency Conversion (CRITICAL-1): Convert acc.initialBalance using CurrencyConverter.convertMinorUnits inside MainViewModel.changeMainCurrency() and invoke accountDao().recalculateAllAccountBalances() within the transaction.
2. Synchronize Reactive UI States (HIGH-1, HIGH-2, HIGH-3): Resolve selectedAccount dynamically from rawAccounts in MainViewModel.uiState; bind AccountsScreen world currency chips directly to state.userPreferences.selectedWorldCurrency; pass weatherEnabled = state.userPreferences.weatherOption != "not_now" into GreetingBar.
3. Align Starting Spent with Custom Billing Cycles (HIGH-4): Key MonthlyPlan.monthKey to the active cycle's start date (cycle.startMillis) rather than the calendar month.
4. Harden Category Detection & Slider Debouncing (MED-1, MED-2, MED-3): Use word-boundary regex matching (\bkeyword\b) in Category.detectFromNote(); trigger DB upserts and audio chimes in onValueChangeFinished on the Analytics budget slider; preserve manual category selection when editing notes in TransactionEditSheet.
5. Complete Bahasa Indonesia Localization & Security Hardening (MED-4, MED-5): Wire remaining hardcoded strings to AppStrings and add a per-install random salt to UserPreferencesRepository.hashPin().